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Spiral conveyor system

Abstract

A spiral conveyor system may include a cage associated with a motor; a conveyor belt traveling helically about the cage; the cage including a plurality of drive elements formed of vertically oriented cage bars; a cage bar cap mounted on at least one of the cage bars; the cage bar cap including a vertically oriented rib extending radially from a surface of the cage bar cap; wherein the rib includes at least one drive face; wherein the conveyor belt includes at least one belt surface configured to engage the at least one drive face; wherein the cage includes a ring that extends between the terminus of the rib and the entrance end of the cage; wherein the surface of the cage bar cap from which the rib extends defines a first diameter; and wherein the ring has a ring diameter that is larger than the first diameter.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application is a continuation of Neely et al., U.S. patent Application Publ. No. 2022/0267098, published on Aug. 25, 2022, and titled “Spiral Conveyor System, which is a continuation of Neely et al., U.S. Patent Appl. Publ. No.

2020/0361713, published on Nov. 19, 2020, and titled "Spiral Conveyor System," which is a continuation of Neely et al., U.S. Patent Appl. Publ. No. 2019/0308817, published on Oct. 10, 2019, and titled "Spiral Conveyor System," which is a continuation of Neely et al., U.S. Patent Appl. Publ. No. 2018/0290833, published Oct. 11, 2018, and entitled "Spiral Conveyor System," which is a continuation of Neely et al., U.S. Patent Appl. Publ. No. 2017/0022012, published Jan. 26, 2017, and entitled "Spiral Conveyor System," which claims priority under 35 U.S.C. § 119 (e) to U.S. Provisional Patent Application No. 62/196,582, entitled "Spiral Conveyor System," and filed on Jul. 24, 2015, which applications are hereby incorporated by reference.

BACKGROUND OF THE INVENTION

(1) The present invention relates to positive drive systems for spiral conveyor belts. In particular, the invention relates to a drum with ribs, where the drive face for engaging the conveyor belt is on the rib and where the rib height above the surface of the drum varies.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The embodiments can be better understood with reference to the following drawings and description. The components in the figures are not necessarily to scale though the scale shown may be considered one embodiment, emphasis instead being placed upon illustrating the principles of the embodiments. Moreover, in the figures, like reference numerals designate corresponding parts throughout the different views.

(2) FIG. 1 shows a schematic diagram of a spiral conveyor system;

(3) FIG. 2 is an enlarged partial perspective view of a portion of a spiral conveyor system showing engagement of a conveyor belt with drive elements of the system;

(4) FIG. 3 is an enlarged partial plan view of a portion of a spiral conveyor system showing engagement of a conveyor belt with drive elements;

(5) FIG. 3A is an enlarged plan view of a portion of a spiral conveyor system showing a portion of a drum and the engagement of different tiers of a conveyor belt with the drum;

(6) FIG. 4 is an enlarged perspective view of a top portion of a spiral conveyor system showing engagement of a conveyor belt with drive elements of the system;

(7) FIG. 4A is a cross-sectional view of a drive element of the system of FIG. 4, taken along line 4A-4A;

(8) FIG. 5 is an enlarged plan view of a portion of a spiral conveyor system showing engagement of a conveyor belt with drive elements that include caps;

(9) FIG. 6 is a partial perspective view of a portion of a spiral conveyor system showing two belt tiers and engagement of the conveyor belt with drive elements;

(10) FIG. 7 is a partial perspective view of a portion of a spiral conveyor system showing two belt tiers and engagement of the conveyor belt with drive elements;

(11) FIG. 8 is an enlarged perspective view of a spiral conveyor system showing drive elements with cage bar caps each having a protruding rib having a drive face to engage with the conveyor belt;

(12) FIG. 9 is a perspective view of a cage bar cap having a protruding rib;

(13) FIG. 9A is a cross-sectional view of the cage bar cap of FIG. 9, taken along line 9A-9A;

(14) FIG. 10 is a perspective view of another embodiment of a cage bar cap having a protruding rib;

(15) FIG. 10A is a cross-sectional view of the cage bar cap of FIG. 10, taken along line 10A-10A;

(16) FIG. 11 is an enlarged perspective view of a spiral conveyor system showing drive elements with integrated protruding ribs;

(17) FIG. 12 is a perspective view of a cage bar cap having an offset and chamfered protruding rib;

- (18) FIG. 12A is a cross-sectional view of the cage bar cap of FIG. 12, taken along like 12A-12A;
- (19) FIG. 13 is a partial enlarged plan view of a conveyor belt engaged with a rib of a cage bar cap;
- (20) FIG. 14 is a partial enlarged plan view of a conveyor belt engaged with an elongated rib of a cage bar cap;
- (21) FIG. 15 is a partial enlarged plan view of a conveyor belt engaged with an elongated rib of a cage bar cap and a support surface;
- (22) FIG. 16 is an enlarged perspective view of the system shown in FIG. 15, a conveyor belt engaged with an elongated rib of a cage bar cap and a support surface;
- (23) FIG. 17 is a partial plan view of a portion of an embodiment of a spiral conveyor system with drive elements with cage bars and cage bar caps, wherein the cage bar caps include protruding ribs;
- (24) FIG. 18 is an enlarged view of another embodiment of a drive element engaged with a conveyor belt;
- (25) FIG. 19 is an enlarged view of an embodiment of a drive element engaged with a conveyor belt showing the clearance between the rib and the belt;
- (26) FIG. 20 is an enlarged view of an embodiment of a drive element engaged with a conveyor belt showing the clearance between the rib and a buttonhead of the belt;
- (27) FIG. 21 is an enlarged plan view of an embodiment of a drive system for a conveyor belt showing an engaged drive element and a disengaged drive element;
- (28) FIG. 22 is an enlarged perspective view of a bottom of a spiral conveyor system showing drive elements with cage bar caps each having a protruding rib having a drive face to engage with the conveyor belt and a smooth belt entrance surface;
- (29) FIG. 23 is a perspective view of an embodiment of a bottom of a spiral conveyor showing the belt engaging with the drive elements having a smooth belt entrance surface;
- (30) FIG. 24 is an enlarged perspective view of an embodiment of an engagement of an edge link of a conveyor belt with a drive element that has a protruding rib and a support surface;
- (31) FIG. 25 is a perspective view of an embodiment of a bottom of a spiral conveyor showing the belt engaging with a ring having a smooth belt entrance surface; and
- (32) FIG. 26 is a perspective view of an embodiment of a top of a spiral conveyor showing a belt disengaging from a drum.

DETAILED DESCRIPTION

(33) For clarity, the detailed descriptions herein describe certain exemplary embodiments, but the disclosure in this application may be applied to any positive drive spiral conveyor system including any suitable combination of features described herein and recited in the claims. In particular, although the following detailed description describes certain exemplary embodiments, it should be understood that other embodiments may be used for positive drive spiral conveyor belts with contoured drive elements.

(34) The spiral conveyor system discussed below is generally a positive drive system, where a drive element comes into direct contact with the conveyor belt to propel the conveyor belt in a travel direction. In the systems discussed below, a central rotating drum includes drive elements that include a drive face for engaging with the conveyor belt. The drive face on the drive element may be adjacent a contoured surface of the drive element, such as a rib that protrudes from the drum or a surface of the drive element. The rib may provide improved geometry for the drive face and also may smooth the travel of the conveyor belt up or down the spiral, particularly when the rib itself has varying height above the drum surface. The conveyor belt may also include a provision to enhance the engagement with the rib, such as a protrusion or a tab with a flat surface for better contact with the drive face.

(35) FIG. 1 shows an embodiment of a spiral conveyor system 1 that may utilize at least one contoured drive element drive system. Spiral conveyor systems such as spiral conveyor system 1 are well known in the art. Spiral conveyor system 1 may include a conveyor belt 15 that is configured to travel a spiral column 5 around a driving drum 10. In some embodiments, driving

drum **10** may include one or more drive elements **20** that engage with conveyor belt **15** for a positive drive system, where the frictional and/or geometric engagement of the drive elements **20** with conveyor belt **15** impart forward motion to conveyor belt **15**. In some embodiments, driving drum **10** and the edge of conveyor belt **15** may include provisions that engage with each other to transfer driving force from the turning drum to the conveyor belt.

(36) Drum **10** may be configured to rotate at various RPM (revolutions per minute), but may be configured to turn at low RPM. The precise speed may depend upon factors such as the height of spiral column **5**, the length of belt **15**, and the intended use of the system, such as to establish a particular cooking, baking, or freezing time. In some embodiments, drum **10** may turn at 15 RPM or less. In some embodiments, drum **10** may turn at 10 RPM or less. In some embodiments, drum **10** may turn at 4 RPM or less. In some embodiments, drum **10** may turn at a rate between 0.1 RPM and 10 RPM, inclusive.

(37) Drum **10** may be rotated using any method known in the art, such as with a motor (not shown) positioned proximate a base **103** of spiral conveyor system **1**. The motor may transfer the power it generates to drum **10** using any mechanism known in the art. In some embodiments, such as the embodiment shown in FIG. **1**, known systems, such as chains and gear boxes to control the transfer of power from the motor to shaft **106**, may be provided. Shaft **106** may be any type of drive shaft known in art, such as an elongated metal pole that extends from base **103** to the top of spiral column **5** along drum central axis **3**. One or more struts such as strut **109** may attach shaft **106** to drum **10** to transfer the rotational force of shaft **106** to drum **10**. Drum **10** may be generally cylindrical in shape and may, in some embodiments, include a cylindrical surface **11** as shown in FIG. **6**. The cylindrical surface may be a continuous cylindrical drum surface formed of sheet metal, or may have a discontinuous surface formed of individual vertical drive elements extending between and connecting circular support bands arranged about the center axis of the drum, or may be a combination of sheet metal and vertical bars as shown in FIG. **1**. Other construction is contemplated to provide a suitable cylindrical surface for guiding an inside edge of belt **15** through spiral column **5**. Drive elements **20** may comprise a contoured rib attached to the sheet metal surface of the drum, such as in the embodiments shown in FIGS. **2-7**, or to a cage bar, as shown in FIGS. **8-26** where the rib extends from a surface of a cage bar cap, or may comprise a combination of a cage bar and a contoured rib. Drive elements **20** may also comprise a cap or covering, such as to provide the contoured surface and/or for improved wear properties, such as shown in FIGS. **4** and **4A**.

(38) A bottom bearing **101** may be provided in, on, or associated with a motor, gearbox, and/or a conveyor frame. As is known in the art, a bottom bearing **101** is provided to reduce rotational friction between the motor and/or base **103** and shaft **106** while supporting radial and axial loads. Bearings and their functionality are generally well known and understood in the art.

(39) Conveyor belt **15** may be modular and include links **25** and connecting rods **26** as shown in FIG. **2**. Conveyor belt **15** may be any type of endless belt known in the art. Conveyor belt **15** may be made from metal, plastic, composites, ceramics, combinations of these materials, or any other type of conveyor belt material known in the art. The particular material may be selected based on factors such as temperature exposure (baking, freezing, room temperature conveying), required tension, length of the belt, ability to clean and/or disinfect the belt, etc. In some embodiments, such as the embodiment shown in FIG. **1**, belt **15** may be an upgoing belt, where belt **15** travels from a bottom **64** of spiral column **5** to a top **63** of spiral column **5**. In other embodiments, the direction of travel may be downgoing, where belt **15** travels from top **63** to bottom **64**. Belt paths at top **63** and bottom **64** may be aligned as shown or arranged at an angle with respect to each other about a central axis **3** of drum **10**.

(40) In use, a conveyor belt **15** enters at one end of drum **10**, typically bottom **64**. Conveyor belt **15** may be fed into the system **1** off of two rollers or sprockets positioned on an axle. Conveyor belt **15** travels through stacked helical tiers around the center drum **10**. Conveyor belt **15** then exits at the

opposite end of drum **10**, typically near top **63**. Conveyor belt **15** may be an endless belt **15**, in which case conveyor belt **15** is fed back to the axle/sprocket at the other end of drum **10** (e.g., in one embodiment, belt **15** travels back to bottom **64** of drum **10** after exiting at top **63** of drum **10**.) In any embodiment, however, the system may be upgoing (traveling from bottom to top) or downgoing (traveling from top to bottom). The gearing and optional weighting of belt **15** at the entrance and/or exit points of spiral **5** may be configured to assist in controlling tension in belt **15** as belt **15** moves through the system **1**. For example, system **1** may include a take-up sprocket **115** configured to pull belt **15** out of spiral column **5**. Take-up sprocket **115** may be located at or after the first terminal roller. Take-up sprocket **115** may be independently driven, such as with a motor **116**. In some embodiments, the motor may be a constant torque motor so that the tension in belt **15** may be controlled within a desired range. In some embodiments, a weighted take-up roller **117** may be provided to assist in maintaining the belt tension along return path **104** and to remove slack belt from the system. Proper tension in belt **15** may inhibit operational issues such as slippage of belt **15** with respect to drum **10**, belt flip-up, or difficulty feeding belt **15** into or out of the helical stack.

(41) In some embodiments, the first tier **80** of belt **15** at the entrance end (e.g., bottom **64**) has a first larger radius and the last tier of belt **15** at the exit end has a second smaller radius. For example, FIG. 3A shows two different tiers, first tier **80** and second tier **90**, of belt **15** made of links **125** and elongated rods **126** that connect the links together. Drum **10** may have a first radius **R1** supporting first tier **80** and a second smaller radius **R2** supporting second tier **90**. First radius **R1** may be at a lower position on drum **10** than second radius **R2**. Belt **15** may be held at either radius by engagement of the circumferential support surfaces of belt **15** with circumferential support surfaces of the drive elements **20**, which are discussed in more detail below with respect to other embodiments.

(42) In some embodiments, belt **15** may have circumferential support surfaces and driving surfaces on the inner edge of belt **15** configured to engage the drive elements. An example of such a belt configuration is shown in FIG. 3 which shows a partial plan view of the embodiment of the spiral system shown in FIG. 2. In FIG. 3, the engagement between the edge link **27** and drive element is clearly shown. Belt **15** may include a protrusion **14** configured to engage the driving face **45** of driving element **20** at engagement point **22**. In some embodiments, protrusion **14** may be a tab or flange extending from an outside surface of edge link **27**. In some embodiments, protrusion **14** may be a portion of a connecting rod **26**. In the embodiment of FIG. 2, driving element **20** comprises a contoured rib that extends between lower support ring **75** and upper support ring **85** and is configured to fit between adjacent belt protrusions **14**. Drive element **20** has a contoured circumferential support surface **23** and a driving face **45** shown in FIG. 3. Due to the direction of travel, the relative position of protrusion **14** and drive element **20** is such that drive element **20** is trapped behind protrusion **14**. Therefore, driving element **20** can push against protrusion **14**. Belt **15** may also include circumferential support surfaces on edge links **27** configured to engage a circumferential support surface of a driving element to hold the belt at a predetermined radius.

(43) FIGS. 4 and 4A show a portion of a top of drum **10**, where each drive element **20** includes a contoured rib **31** and a rib cover or wear cap **41**. In the embodiment shown in FIG. 4, wear cap **41** follows the contoured surface of contoured rib **31** and provides drive face **45** shown in FIG. 3 for engaging conveyor belt **15**. In the embodiment shown in FIG. 4, cage bar caps **40** are separate and distinct from contoured ribs **31** and are attached to contoured ribs **31**. In some embodiments, the attachment may be fixed and permanent, such as by welding, using adhesives, or with rivets. In other embodiments, the attachment may be removable, such as when press-fitted, interference-fitted, with screws, clips, or the like.

(44) FIG. 4A, a cross-section of a driving element from FIG. 4, shows that contoured rib **31** is distinct from wear cap **41**. As shown in FIG. 4A, contoured rib **31** is solid in cross-section and wear cap **41** is a three-sided, U-shaped portion of material that is sized and shaped to fit over and partially surround contoured rib **31** so that wear cap **41** faces outward towards belt **15**. In other

embodiments, contoured rib **31** may be hollow or have any other cross-sectional shape. As shown in FIG. **4**, wear cap **41** is oriented on contoured rib **31** so that wear cap **41** faces conveyor belt **15**. In this embodiment, wear cap **41** may be attached to contoured rib **31** using any of the methods discussed above. In this embodiment, wear cap **41** may act as a lower coefficient of friction cover of contoured rib **31** and simply follow the contours of contoured rib **31**. Although shown as generally parallel to the belt edge, contoured surface **23** may be angled or have a chamfered edge, and wear cap **41** may be correspondingly angled with a chamfered edge to improve engagement with the belt.

(45) Contoured rib **31** may be made of any material, such as metal, plastic, or composite. Wear cap **41** may be made of the same material as contoured rib **31** or a different material. For example, in a typical embodiment, contoured rib **31** may be made from a metal while wear cap **41** may be made from a plastic material. In other embodiments, contoured rib **31** may be made from a plastic material and wear cap **41** may be made from the same plastic material. In other embodiments, contoured rib **31** may be made from a plastic material while wear cap **41** may be made from a different plastic material. Any combination of materials may be appropriate depending upon the intended use. For example, both contoured rib **31** and wear cap **41** may be made from the same metal or metals having similar thermal expansion properties for baking embodiments. Similarly, both contoured rib **31** and wear cap **41** may be made from the same plastic material or plastic materials having similar low temperature properties for freezing embodiments. Further, wear cap **41** may be made from a material with a lower coefficient of friction and/or more durable wear properties than contoured rib **31** for long-term engagement with conveyor belt **15**. In some embodiments, wear cap **41** may be a sacrificial wear component.

(46) FIG. **5** shows a partial plan view of the embodiment of the spiral system shown in FIG. **4**. In FIG. **5**, the engagement between the edge link **27** and wear cap **41** is more clearly shown. Wear cap **41** follows a contoured surface of contoured rib **31**, creating a circumferential support surface **46** on wear cap **41** for contacting corresponding outer edge surface **28** of edge link **27** belt **15**. Wear cap **41** also covers a driving face of contoured rib **31**, creating a driving face **45** on wear cap **41** for engaging protrusion **14** of belt **15**. Due to the direction of travel, the relative position of protrusion **14** and wear cap **41** is such that wear cap **41** is trapped behind protrusion **14**. Wear cap **41**, and, therefore, driving element **20** can push against protrusion **14**.

(47) Contoured surface **23** shown in FIG. **4** may have any combination of curved, tapered, and flat portions. In some embodiments, contoured surface **23** varies the height of the drive face above drum surface **11**. This height variation allows for improved tension control as belt **15** travels along spiral tower **5**. In some embodiments, contoured surface **23** is arranged so that as belt **15** follows the contour, the height either remains constant or decreases—the height does not increase in the direction of belt movement. This constant or decreasing height of contoured surface **23** above drum surface **11** serves to prevent increases in belt tension that would otherwise result from increasing the radius of a tier of belting.

(48) FIG. **6** shows two tiers of an edge driven system with contoured drive elements showing the relative position of the two tiers at different heights on spiral tower **5**. In this embodiment, drive element **20** is a contoured rib attached to drum **10**, where drive element **20** includes a contoured surface **23** that has three sections of varying height above drum surface **11**: a lowest section **91** that extends the furthest away from drum surface **11**, an upper section **92** whose surface is closer to drum surface **11** than a surface of lowest section **91**, and a tapered section **93** that connects lower section **91** and upper section **92**. In this embodiment, lower section **91** and upper section **92** are both flat, with unvarying height above drum surface **11** within the discrete section. In other embodiments, any section may include subsections with varying height and/or continuous tapers like tapered section **93**.

(49) Contoured surface **23** positions belt **15** at different radii depending upon the position of belt **15** on drum **10**. First tier **80** is positioned proximate a bottom **64** of drum **10** on lower section **91** of

driving element **20**. Second tier **90** is positioned proximate top **63** of drum **10** on upper section **92** of driving element **20**. Lower section **91** is the highest part of contoured surface **23** while upper section **92** is closer to drum surface **11** than lower section **91**. Therefore, first tier **80** is positioned further away from drum surface **11** than second tier **90**. The loop of first tier **80** around drum **10** has a greater radius than that of second tier **90**. As a weighted roller often assists in the take-up of belt **15** when exiting spiral **5** as shown by weighted roller **117** in FIG. **1**, the smaller radius proximate the spiral exit may help to produce a constant tension in belt **15** in all tiers of spiral **5**. FIG. **7** shows a similar arrangement of first tier **80** and second tier **90** but on drive elements **20** that have a constant taper in the lower portion that is furthest away from drum surface **11** at bottom **64** and a constant height in the upper portion closest to drum surface **11** at exit point **67**.

(50) The embodiments shown above with respect to FIGS. **2-7** show embodiments that may be particularly suitable for high temperature uses, such as cooking and baking. Oven and/or cooker drums are generally constructed as cages, such as is best shown in FIG. **4**, or as continuous sheet metal cylinders, such as is best shown in FIGS. **6** and **7**. While rapid wear due to contact between the drum and belt is typically tolerated, some expensive high temperature plastics such as PEEK (poly ether ether ketone) may be used to increase the life of the parts. However, the high costs and more rigid mechanical properties may make PEEK and similar materials undesirable. Therefore, while any material may be appropriate for the drive components, all drive components in the embodiments shown in FIGS. **2-7** may be made from metals to be cost effective and relatively easy to machine/manufacture. However, in an embodiment such as the embodiment shown in FIG. **4**, a relatively small wear cap **41** is provided. Wear cap **41** has a much smaller cross-sectional area than a standard cage bar cap, such as the cage bar cap **140** shown in FIGS. **9** and **9A** below. As such, a small wear cap like wear cap **41** may be cost-effective to produce in PEEK or a similar high temperature material.

(51) In contrast, the embodiments shown below with respect to FIGS. **8-26** may be manufactured for lower temperature systems, such as for any use at less than about 100 degrees C., for example room temperature conveying or freezing operations. In such embodiments, a wider variety of materials may be used for the drive surfaces. In many embodiments, a drum may be constructed in a typical cage configuration, with upper and lower rings connected by vertical cage bars, such as shown in FIG. **8**. In cage bar embodiments, the cage bars may be covered with cage bar caps made from inexpensive materials for use as sacrificial wear components. A typical material is UHMW (ultra high molecular weight polyethylene). Such inexpensive wear materials may not be appropriate for higher temperature applications due to glass transition and melt temperatures. However, for lower temperature systems, UHMW and similar materials are easily extruded, even when including a protruding and, in some embodiments, contoured rib. Machining such a rib is also readily achieved. These lower cost manufacturing abilities make such fabrications even lower cost than forming similar structures from metals such as steel. Further, because UHMW and similar materials may be somewhat soft, cage bar caps made from UHMW may be readily manipulated to snap onto a cage bar. Finally, such UHMW cage bar caps may be used to retrofit existing systems with ribbed cage bar caps by simply replacing an existing ribless cage bar cap with a ribbed cage bar cap, particularly when installing the ribbed cage bar in a freezer or other hard-to-access system where the cage bar may be bent to be maneuvered into the proper position.

(52) In some embodiments, contoured surface **23** may also include additional geometry to better engage belt **15**. FIGS. **8-26** show various embodiments of drive elements **20** that include a rib that protrudes from the surface of drive elements **20** toward belt **15**. These ribs include the height varying contoured surface **23** and also provide a drive face configured to contact a portion of belt **15**, such as a tab on an edge link as discussed above.

(53) FIG. **8** shows a partial perspective view of a drum **8010** that includes drive elements **8020**. Rib **8021** is a protrusion that extends away from a base surface **8032** of cage bar cap **8040** towards belt **8015** and establishes a contoured surface **8023**. Similar to contoured surface **23** discussed above

with respect to FIG. 6, contour surface **8023** includes a flat lower section **8041** at a first height above base surface **8032**, an upper section **8043** at a second, lower height above base surface **8032**, and a tapered section **8047** that continuously joins lower section **8041** to upper section **8043**. A leading edge **8046** of rib **8021** may include a drive face **8045** that engages with belt **8015**. In some embodiments, belt **8015** may engage the drive face **8045** with the end surface **8028** of a link leg **8018**.

(54) FIGS. 9 and 9A show another embodiment of a cage bar cap **140** with an offset rib **121** that may be used in a spiral system such as the system shown in FIG. 8. In this embodiment, cage bar cap **140** is a structure configured to fit onto cage bar **8030** and includes a body with four walls that define an interior cavity **143**: a first wall **130**, a second or front wall **132**, third wall **133**, and an open, discontinuous fourth wall that includes a fourth wall first portion **134** and a fourth wall second portion **135**. Cavity **143** is configured to receive a cage bar (not shown), so that first wall **130**, second wall **132**, third wall **133**, and fourth wall portions **134**, **135** cover the cage bar. While first wall **130**, second wall **132**, third wall **133** may fully cover a corresponding cage bar wall, discontinuous fourth wall portions **134**, **135** will only cover a portion of a corresponding cage bar wall, as a gap **136** between fourth wall first portion **134** and fourth wall second portion **135** allows a cage bar to be inserted into cavity **143** or allows for some give in cage bar cap **140** to permit cage bar cap **140** to slide onto a cage bar. Cage bar cap **140** may be made of any metal or synthetic material.

(55) Cage bar cap **140** may be oriented on a cage bar so that front wall **132** is configured to face a conveyor belt (not shown) when cage bar cap **140** is installed in a spiral system.

(56) Cage bar cap **140** includes a rib that extends away from the body. In this embodiment, offset rib **121** protrudes away from a base surface **142** of front wall **132** to a rib height, and a length of offset rib **121** is coextensive with a top-to-bottom length of cage bar cap **140**. In this embodiment, rib height varies along the length of offset rib **121**. Similar to embodiments discussed above, offset rib **121** includes three continuous portions: an upper section **110**, a lower section **111**, and a tapered section that connects upper section **110** and lower section **111**. Upper section **110** extends a first height **151** above cage bar cap surface **142**, where first height **151** is constant; tapered section **112** extends a second height **152** above cage bar cap surface **142**, where second height varies along the length of tapered section **112**; and lower section **111** extends a third height **150** above cage bar cap surface **142**. In this embodiment for an upgoing belt, third height **150** is greater than first height **151**, while second height **152** tapers from third height **150** to first height **151**. In other embodiments where the belt is downgoing, offset rib **121** may be inverted.

(57) Third height **150** may be based on several factors, including the size of the drum, the tension of the incoming belt, and the elasticity of the belt. Third height **150** may be chosen to create an increase in the circumference of a first tier of belting approximately equal to or a significant percentage of the amount of stretch in the belt. In this way, when the belt moves to a position around the second lower rib height, there will be very little stretch, and hence tension, remaining in the belt.

(58) First height **151** may be determined by the height of any edge features for engaging with the drive cap, such as protrusions **14** shown in FIG. 5. In some embodiments, offset rib **121** may be designed so that an end of an edge feature is configured to contact the face **142** of the cage bar cap **140**. In other embodiments, offset rib **121** may be designed so that an end of an edge feature does not contact face **142** of cage bar cap **140**. This choice is typically based on wear considerations. For example, the end of an edge feature on a metal link may be rough or sharp as a result of the link being punched from sheet metal material. In this case, it may be advantageous for offset rib **121** to support the link away from face **142** and prevent the rough end surface of an edge feature from wearing face **142** of cage bar cap **140**. In the case of a molded plastic link, it may be preferable to design a flat wear surface on the end of the protrusion specifically to contact face **142** of the cage bar cap **140** to position a link or prevent wear on a tension-bearing portion of the link.

(59) Taper angle **125** may be restricted by factors such as the belt tension, belt weight, and friction coefficient between the belt and cage bar cap. Taper angle **125** may be limited so that the radial force created by the belt tension is inhibited from causing an inside edge of the belt to lift off the support rails and move upward along the taper toward a portion of the drum with a smaller radius. In this regard, higher belt weight and higher coefficient of friction will also inhibit such movement. A minimum length of the taper may be determined by the difference in height between the first height **151** and second height **150** and the maximum taper angle **125** that will inhibit lifting the inside edge of the belt. However, the length of tapered section **112** can be as long as desired and may be as long as the offset rib **121** itself, e.g., offset rib **121** may have a continuous taper along a length of offset rib **121** so that offset rib **121** may have a continuously variable height.

(60) In some embodiments, third height **150** may be between one-half ($\frac{1}{2}$) inch to two (2) inches. First height may be between $\frac{1}{4}$ inch to $\frac{3}{8}$ inches. Tapered section **112** may taper at an angle **125** that is 30 degrees or less from third height **150** to first height **151**. In other embodiments, any of these heights may be greater or lower than these ranges.

(61) Offset rib **121** has a thickness **147**. In this embodiment, thickness **147** is constant along the entirety of offset rib **121**. Thickness **147** may be selected based on a number of factors, but is generally proportional to the rib height; when height of offset rib **121** increases, so does thickness **147**. In such embodiments, the height-thickness proportion may be selected based upon the highest rib height, such as third height **150** in the embodiment shown FIG. **9**. In some embodiments, offset rib **121** is structurally rigid enough to drive the belt without significant deflection, which may be considered to be more than approx. 5° from base surface **142**, i.e. remaining at an angle that is 90° to 95° with respect to the face of cage bar cap). However in some embodiments, a positive drive system can operate with a drive face that is not perpendicular and is angled as much as 45° with respect to the face of the cage bar cap (i.e., ranging from 45 degrees to 135 degrees with respect to the cage bar cap), depending on belt tension, friction, etc. In such embodiments, any drive face may be correspondingly angled (as is known in the art, sprocket and gear faces are generally angled or curved in this way). In those embodiments where a conveyor belt includes edge features for engaging with the drive cap, such as protrusions **14** shown in FIG. **5**, thickness **147** may be limited by any spacing between those belt edge features so that thickness **147** does not exceed that interstitial spacing. This spacing may vary as the belt may be collapsible to accommodate the curves of the spiral. In such cases, thickness **147** may be selected to accommodate the minimum spacing—the spacing between edge features when the belt is fully collapsed. Similar thickness restrictions may be placed on any rib discussed herein, including drive element **20** discussed above with respect to FIGS. **2-7**.

(62) In the embodiment shown in FIGS. **9** and **9A**, offset rib **121** is positioned proximate third wall **133**. As such, offset rib **121** is shifted away from a central axis **100** of cage bar cap **140** an offset **D2** and is positioned a first distance **D1** away from leading face **130**. This offset may be beneficial in stiffening offset rib **121** due to additional support from third wall **133**. Additionally, because offset rib **121** is essentially a smooth extension of third wall **133**, a crevice is eliminated, which may inhibit an accumulation of debris on cage bar cap **140** and/or may make cage bar cap **140** easier to clean. It may also be easier to machine the contour of offset rib **121** by laying third wall **133** flat on a machining surface.

(63) FIGS. **10** and **10A** show a cage bar cap **240** that is essentially the same as cage bar cap **140**, except that central rib **221** is aligned with central axis **200**. In such symmetrical embodiments, the symmetry of cage bar cap **40** reduces necessary orientation during manufacturing, such as machining or extruding cage bar cap **140** and central rib **221**. The symmetry also reduces part inventory as pre-machined parts may be used on either clockwise or counter-clockwise rotating systems.

(64) An advantage common to the embodiments shown in FIGS. **9-10a** is that these embodiments can be made easily and inexpensively, such as by extruding profiles corresponding to the cross

sectional areas of FIGS. **9A** and **10A**. Extruded profiles can then be cut to length and the ribs machined to produce the desired contoured surfaces.

(65) In some embodiments, the cage bar itself may include a rib as opposed to separate cage bars and cage bar caps. For example, FIG. **11** shows an embodiment of a positive drive spiral conveyor system **301** that includes an offset rib **321** having a varying height contour surface **323** that is similar to offset rib **121** shown in FIG. **9**. However, unlike FIG. **9**, offset rib **321** extends away from a surface **342** of cage bar **330** instead of from a cage bar cap. Cage bar **330** is uncapped and attached to drum **310**. In this embodiment, cage bar **330** and offset rib **321** form a single monolithic structure. System **301** otherwise is similar to or the same as any of the systems discussed above.

(66) FIGS. **12** and **12A** show another embodiment of a cage bar cap **440** configured to provide a contoured surface **423**. Like the embodiment of cage bar cap **140** shown in FIG. **9** in most respects, cage bar cap **440** is configured to cover a cage bar (not shown). However, in this embodiment, cage bar cap **440** has some geometrical variations. Cage bar cap **440** includes a short offset rib **421**. Short offset rib **421** is similar to offset rib **121**, as short offset rib **421** extends away from a surface **442** of cage bar cap **440** offset rearward from a central axis **400** a distance **452** and a second distance **450** from leading face **430**. Short offset rib **421** also varies in height along a height of cage bar cap **440**.

(67) Unlike offset rib **121**, short offset rib **421** is not coextensive with cage bar cap **440**. Short offset rib **421** extends from a top **463** of cage bar cap **440** to a terminus **424**. Terminus **424** is separated from a bottom **464** of cage bar cap **440** by a distance **410** so that short offset rib **421** terminates short of the position at which conveyor belt **415** enters the spiral. This allows cage bar cap **440** to have a smooth bottom portion **497**. Smooth bottom portion **497** may be beneficial in guiding a conveyor belt onto contoured surface **423** after allowing the links of the belt to collapse. Distance **410** may be any distance desired, but may be less than 25% of length of cage bar cap **440**. Smooth bottom portion **497** may be wider than the rest of cage bar cap **440**. In such embodiments, an angled transition portion **498** connects smooth bottom portion **497** with surface **442** to avoid possible jarring lateral motion of the belt as the belt engages short offset rib **421** and also to eliminate a possible niche which might be difficult to clean.

(68) Also unlike offset rib **121**, short offset rib **421** includes a chamfered edge **432**. Chamfered edge **432** is positioned on an opposite side of short offset rib **421** to a drive face **445**. Chamfered edge **432** may be provided to facilitate movement of the belt along short offset rib **421** and/or to facilitate manufacturing. Chamfered edge **432** extends away from trailing face **433** of cage bar cap **440** at an angle **430**. Angle **430** may be any angle desired, but may in some embodiments be between 20 degrees and 90 degrees.

(69) FIGS. **13** and **14** show an embodiment of how a cage bar cap **540** on a cage bar **530** may engage with a conveyor belt **515**. For clarity, the drum is not shown, though cage bar **530** would be attached to a drum. Cage bar **530** and cage bar cap **540** is similar to any cage bar cap discussed above, though in this embodiment center rib **521** is centrally located on a front face **532** of cage bar cap **540**. Center rib **521** includes a drive face **545** configured to engage with a portion of an edge link **527** of a metal belt **515**.

(70) In FIG. **13**, metal belt **515** may include a plurality of generally U-shaped links **525** connected by rods **526** in any known manner that permits metal belt **515** to expand and collapse as it moves through a spiral. In this embodiment, all links **525** are edge links that form the outermost surfaces of metal belt **515**. In this embodiment, edge links **527** include tabs **514** configured to engage with drive face **545**. As shown, drive face **545** abuts tab **514** to firmly and yet removably engage metal belt **515** while tab **514** may touch or be spaced apart from surface **532**. As discussed above with respect to FIG. **9**, a thickness of center rib **521** may be limited by an interstitial space **555** between adjacent tabs of adjacent edge links **527**, as center rib **521** is designed to fit easily into interstitial space **555**. In some embodiments, center rib **521** may extend to edge link **527**, while in other embodiments, center rib **521** may terminate short of edge link **527**.

(71) Tab **514** extends away from outer leg **518** of an edge link **527** at a tab angle **516**. Tab angle **516** may be any desired angle, but may be between 20 degrees and 130 degrees. Tab **514** may include a flat face to provide a larger surface area for the engagement of drive face **545** with tab **514**. In such embodiments, tab angle **516** may be about 90 degrees. In this embodiment, edge links **527** are made from a metal material. As such, tab **514** may be formed on outer leg **518** by bending a length of outer leg to the desired angle. Other methods of manufacturing such an edge feature are also contemplated, such as stamping edge link **527** into the desired shape.

(72) FIG. **14** shows the embodiment of FIG. **13**, but at a different portion of the rib where the rib portion **621** is at a greater height above surface **532** of cage bar cap **540** on a cage bar **530** of a driving element **520** may engage with a conveyor belt **515**. For clarity, the drum is not shown, though cage bar **530** would be attached to a drum. Cage bar **530** and cage bar cap **540** are similar to any cage bar cap discussed above, though in this embodiment extended center rib portion **621** is centrally located on a front face **532** of cage bar cap **640**. Center rib portion **621** includes a drive face **645** configured to engage with a portion of an edge link **527** of a metal belt **515**.

(73) Extended center rib portion **621** has a rib height **650** that is higher than a rib height of center rib **521** and the length of tab **514**. This greater height allows for greater clearance between tab **514** and front face **532** for those embodiments where contact between tab **514** and front face **532** is not desired, such as when, for example, tab **514** may have a sharp edge that would increase wear of front **532** and potentially limit a lifespan of cage bar cap **540**.

(74) In some embodiments, such as the embodiment shown in FIGS. **15** and **16**, a support structure **773** is provided adjacent to a guide rib **721**. In this embodiment, a cage bar **730** and a cage bar cap **740** of a drive element **720** are the same as or similar to any cage bar or cage bar cap discussed above, particularly those with centrally positioned ribs such as cage bar cap **540**. In this embodiment, a tabbed belt **715** is similar to metal belt **615** as discussed above, with edge links **727** having drive tabs **714**.

(75) Support structure **773** is sized and positioned so that when a drive face **745** is engaged with a first surface of drive tab **714** of an edge link **727**, support structure **773** contacts a second surface of drive tab **714** at an engagement point **742** on support structure **773**. In the embodiment shown in FIGS. **15** and **16**, the first surface is substantially perpendicular to the second surface of drive tab **714**. This engagement may lend additional stability to tabbed belt **715** as tabbed belt **715** travels through the spiral.

(76) As shown in FIG. **16**, support structure **773** may follow the geometry of guide rib **721** though support structure **773** does not extend as far above cage bar surface **732** as does guide rib **721**. This configuration is so that engagement tab **714** may be nestled within the L-shape formed by guide rib **721** and support structure **773**. For example, in those embodiments where guide rib **721** includes a tapered portion **747**, support structure **773** also includes a taper.

(77) As shown in FIG. **17**, ribbed guide structures are not limited to embodiments with metal belts. FIG. **17** shows a plastic link belt **815** that is formed from a plurality of plastic links **825** joined by elongated rods **826**. Elongated rods **826** may be made from metal or plastic. In this embodiment, a cage bar **830** and a cage bar cap **840** of a drive element **820** are the same as or similar to any cage bar or cage bar cap discussed above, particularly those cage bar caps with centrally positioned ribs such as cage bar cap **540**.

(78) In this embodiment, each pitch includes an edge link **827** from which a tab **814** protrudes towards an engagement rib **821**. Tab **814** is similar to protrusion **14** as discussed above. Tab **814** may engage with engagement rib **821** in any manner discussed above. As shown in FIG. **17**, multiple drive elements **820** may be engaged with different corresponding edge links **827** simultaneously.

(79) FIG. **18** shows how a chamfered rib **1021** may engage with a metal belt **1015**. For clarity, the drum is not shown, though cage bar **1030** would be attached to a drum. Cage bar **1030** and cage bar cap **1040** are similar to any cage bar and cage bar cap, respectively, discussed above. In this

embodiment chamfered rib **1021** is centrally located on a front face **1032** of cage bar cap **1040**. Chamfered rib **1021** includes a drive face **1045** configured to engage with a portion of an edge link **1027** of a metal belt **1015**.

(80) In this embodiment, metal belt **1015** may include a plurality of generally U-shaped links **1025** connected by rods **1026** in any known manner that permits metal belt **1015** to expand and collapse as it moves through a spiral. In this embodiment, all links **1025** are edge links **1027** that form the outermost surfaces of metal belt **1015**. In this embodiment, edge links **1027** include tabs **1014** configured to extend away from an edge link **1027** and engage with drive face **1045**. As shown, drive face **1045** abuts tab **1014** to firmly and yet removably engage metal belt **1015**.

(81) Tab **1014** may include a flat face to provide a larger surface area for the engagement of drive face **1045** with tab **1014**. In this embodiment, edge links **1027** are made from a metal material. As such, tab **1014** may be formed by bending a length of an outer leg of edge link **1027** to a desired angle. Other methods of manufacturing such an edge feature are also contemplated, such as stamping edge link **1027** into the desired shape.

(82) Because chamfered rib **1021** has a chamfered surface **1075**, chamfered rib **1021** essentially terminates at a point **1044**. Point **1044** is thin enough to slide past tab **1014** and into a niche **1029** between tab **1014** and an adjacent edge link **1027A** when tab **1014** is engaged with drive face **1045**. Such an engagement may be more stable and secure than ribs that lack a chamfered edge or other point-like terminal ends. Chamfered surface **1075** may be angled to correspond to an angled portion **1090** of edge link **1027**. Chamfered surface **1075** may permit separation of edge links **1027** and **1027A** such that point **1044** may contact link surface **1091**. As will be recognized by those in the art, any chamfered surface of any embodiment disclosed herein or adapted to have a chamfered surface may also serve as the contoured surface or a portion of the contoured surface.

(83) In some embodiments, conveyor belts may include edge features other than edge features for engaging with a positive drive system. For example, as shown in FIG. **19**, a buttonless belt **1115** may include welds **1192** that cover the ends of elongated rods **1126**. While welds **1192** may be smooth, welds **1192** may protrude away from an outer leg **1118** of an edge link **1127**. Edge link **1127** may also include an engagement tab **1114** similar to protrusion **14** or any other tab described above, particularly tab **114**, an angled portion of an outer leg of an edge link.

(84) A chamfered rib **1121** of cage bar cap **1140** may engage with buttonless belt **1115**. Cage bar **1130** and cage bar cap **1140** is similar to any cage bar cap discussed above. In this embodiment chamfered rib **1121** is similar to chamfered rib **1021** discussed above and is centrally located on cage bar cap **1140**. In some embodiments, rib **1121** may be sized and dimensioned so that rib **1121** has a clearance gap **1195** between chamfered edge **1183** and welds **1192** as rib **1121** moves towards an engagement position with an engagement tab **1114**. This arrangement may prevent or inhibit unwanted motion that would otherwise occur if rib **1121** were to come into contact with weld **1192**. This arrangement may also prevent engagement with weld **1192**, which may undesirably wear either or both of rib **1121** and weld **1192**.

(85) FIG. **20** shows a similar clearance when a chamfered rib **1283** engages with a buttonhead belt **1215**. Buttonhead belt **1215** may include buttonheads **1293** that cover the ends of elongated rods **1226**. While buttonheads **1293** may be smooth, buttonheads **1293** may protrude away from an outer leg **1218** of an edge link **1227**. Edge link **1227** may also include an engagement tab **1214** similar to protrusion **14** or any other tab described above, particularly tab **124**, an angled portion of an outer leg of an edge link.

(86) A chamfered rib **1221** of cage bar cap **1240** may engage with buttonhead belt **1215**. Cage bar **1230** and cage bar cap **1240** is similar to any cage bar cap discussed above. In this embodiment chamfered rib **1221** is similar to chamfered rib **1021** discussed above and is centrally located on cage bar cap **1240**. In some embodiments, chamfered rib **1221** may be sized and dimensioned so that rib **1221** has a clearance gap **1295** between chamfered edge **1283** and buttonheads **1293** as chamfered rib **1221** moves towards an engagement position with an engagement tab **1214**. This

arrangement may prevent or inhibit unwanted motion that would otherwise occur if chamfered rib **1221** were to come into contact with buttonhead **1293**. This arrangement may also prevent engagement with buttonhead **1293**, which may undesirably wear either or both of chamfered rib **1221** and buttonhead **1293**.

(87) FIG. **21** shows an embodiment of how drive elements **1320** including a cage bar **1330** and a cage bar cap **1340** on a cage bar **1330** may engage with a conveyor belt **1315**. Cage bar **1330** is attached to a drum **1310**. Drum **1310**, cage bar **1330**, and cage bar cap **1340** are similar to any drum, cage bar, and cage bar cap, respectively, discussed above, though in this embodiment rib **1321** is centrally located on a front face of cage bar cap **1340**. Chamfered center rib **1321** includes a drive face **1345** configured to engage with a portion of an edge link **1327** of a metal belt **1315**.

(88) In this embodiment, metal belt **1315** may include a plurality of generally U-shaped links **1325** connected by rods **1326** in any known manner that permits metal belt **1315** to expand and collapse as it moves through a spiral. In this embodiment, edge links **1327** include tabs **1314** configured to engage with drive face **1345**. As shown, rib **1321** abuts tabs **1314** to firmly and yet removably engage metal belt **1315**. Tabs **1314** extend away from outer leg **1318** of an edge link **1327** and are similar to tabs **514** discussed above with respect to FIG. **13**.

(89) FIG. **21** shows how a top tier **1390** of a spiral may exit the spiral and head to a return path, such as return path **104** as shown in FIG. **1**. In FIG. **21**, first drive element **1320** is engaged with a tab **1314**, while second drive element **1320A** is disengaging from tab **1314A**. At the point of disengagement, links **1325** expand from a first collapsed pitch **P1** to a second expanded pitch **P2**, resulting in forward movement of links **1325** with respect to ribs **1321** and **1321A**. The sharp angle of chamfered edge **1383A** results in no rib material blocking or inhibiting movement of tab **1314B**. In some embodiments, tab **1314B** may even slide along chamfered edge **1383** for a smooth, not sudden, movement of tab **1314B** past rib **1321A**. Sudden catching and disengagement of the tabs from the drive elements may damage the drive elements and place unintended forces on the belt which may reduce the useful life of the belt. In some extreme cases, a sudden disengagement may derail the conveyor belt and/or jar conveyed product in undesirable ways.

(90) Similarly, a jarring engagement with the drum at a spiral entrance may produce undesirable results. FIGS. **22-24** show how a smooth bottom portion **1497** can help to ease belt **1415** onto a rib **1421**. A drum **1410**, which is similar to any drum discussed above, includes multiple drive elements **1420** with contoured surfaces **1423**. Rib **1421** is a protrusion that extends away from a base surface of drive element **1420** towards belt **1415**. Similar to contoured surface **23** discussed above with respect to FIG. **6**, contour surface **1423** includes a flat lower section **1448** at a first height above the base surface of drive element **1420**, an upper section **1446** at a second, lower height above the base surface of drive element **1420**, and a tapered section **1447** that continuously joins lower section **1448** to upper section **1446**.

(91) In this embodiment, drive element **1420** may have a smooth bottom portion **1497** that is similar to smooth bottom portion **497** discussed above. Smooth bottom portion **1497** may be beneficial in guiding a conveyor belt onto contoured surface **1423** after allowing the links to collapse. In some embodiments, smooth bottom portion **1497** may be wider than the rest of drive element **1420**. In such embodiments, an angled transition portion **1487** connects smooth bottom portion **1497** with an outermost surface of lower section **1448** to avoid possible jarring motion of the belt as the belt connects with rib **1421**, as belt **1415** has an opportunity to first transition from being frictionally driven by drive element **1420** before beginning the trip up the spiral on rib **1421** from spiral bottom **1464** to spiral top **1463**. In this embodiment, that transition is further eased because bottom portion **1497** has a greater diameter than lower section **1448**, where the additional tension in a more expanded belt being held at a greater diameter can release onto the lower diameter rib **1421** as belt **1415** moves from a first position **T1** to a second position **T2**. In FIG. **24**, bottom portion **1497** has essentially the same or a slightly smaller diameter than flat lower section **1448** of rib **1421**. Due to the overlapping engagement of the link tabs with lower section **1448**, belt

1415 at **T2** may have the same or slightly smaller diameter than belt **1415** at **T1**. A chamber at the bottom of section **1448** provides a smooth transition from bottom portion **1497** to flat lower section **1448**. In such embodiments, lateral movement is minimized as links **1425** engage ribs **1421**.

(92) In another embodiment, shown in FIG. 25, instead of a flat bottom portion of a drive element providing the larger diameter, smooth surface for onboarding to a rib, drum **1510** includes a lower ring **1598** that is approximately the same diameter as a lower portion **1548** of a rib **1521**. Rib **1521** is associated with a drive element **1520** of drum **1510**. Rib **1521** includes a flat lower section **1548** at a first height above the base surface of drive element **1520**, an upper section **1546** at a second, lower height above the base surface of drive element **1520**, and a tapered section **1547** that continuously joins lower section **1548** to upper section **1546**. Drum **1510**, drive element **1520**, and rib **1521** may have the same features as any drum, drive element, or rib, respectively, discussed above. Ring **1598** performs the same or a similar function as bottom portion **1497** discussed above by allowing belt **1515** to collapse on smooth ring **1598**, then drum **1510** advances belt **1515** on to positively engage rib **1521**.

(93) FIG. 26 shows an embodiment of a drum **2010** that includes a similar ribless portion **2041** of a drive element **2020** proximate a top **2063** of system **2001**. In this embodiment, drum **2010**, drive element **2020**, cage bar cap **2040**, and rib **2021** may be similar or the same as any drum, drive element, cage bar cap, and rib discussed above. Similar to rib **1521**, rib **2021** includes a flat lower section **2048** at a first height above the base surface of cage bar cap **2040**, an upper section **2046** at a second, lower height above the base surface of cage bar cap **2040**, and a tapered section **2047** that continuously joins lower section **2048** to upper section **2046**.

(94) Proximate bottom **2064**, drive bar cap **2040** includes a smooth bottom portion **2097** that is similar in form and function to smooth bottom portion **1497** as discussed above. Ribless portion **2041** performs a similar function proximate top **2063**. In embodiments where rib **2021** terminates at an upper point **2095** on cage bar cap **2040**. Upper point **2095** is separated from a cage bar top **2042** of cage bar cap **2040** so that rib **2021** terminates short of the position at which conveyor belt **2015** exits the spiral. Ribless portion **2041** allows for belt **2015** to expand unhindered. This may allow for more tension control of belt **2015** as belt **2015** exits the spiral and is pulled by a take up reel such as take up roller **115** shown in FIG. 1, which may inhibit slippage of belt **2015**.

(95) While various embodiments have been described, the description is intended to be exemplary, rather than limiting, and it will be apparent to those of ordinary skill in the art that many more embodiments and implementations are possible that are within the scope of the embodiments. Any feature of any embodiment may be used in combination with or as a substitute for any other feature or element in any other embodiment unless specifically restricted. Accordingly, the embodiments are not to be restricted except in light of the attached claims and their equivalents. Also, various modifications and changes may be made within the scope of the attached claims.

Claims

1. A spiral conveyor system comprising: a drum associated with a motor, wherein the drum extends from a drum bottom to a drum top, and includes a plurality of vertically oriented drive elements extending from the drum bottom to the drum top; and a conveyor belt traveling helically about the drum from an entrance end of the drum to an exit end of the drum; wherein the drive element includes a vertically oriented rib extending radially from a surface of the drum; wherein the rib includes at least one drive face; wherein the conveyor belt includes at least one belt surface configured to engage the at least one drive face; wherein a height of the rib above of the surface of the drum varies along a length of the rib; and wherein the rib is positioned off center on the drive element.
2. The spiral conveyor system of claim 1, wherein the rib has a maximum height of the rib at one end of the rib.

3. The spiral conveyor system of claim 1, wherein at least a portion of the rib has a tapered height.
4. The spiral conveyor system of claim 1, wherein the rib further includes a chamfered surface configured to engage with a lateral portion of the conveyor belt.
5. The spiral conveyor system of claim 4, wherein the chamfered surface angled facing partially in a rearward direction opposite the direction of belt travel about the drum.
6. The spiral conveyor system of claim 5, wherein the rib is disposed flush with a trailing edge of the drive element.
7. A spiral conveyor system comprising: a drum associated with a motor, wherein the drum extends from a drum bottom to a drum top, and includes a plurality of vertically oriented drive elements extending from the drum bottom to the drum top; and a conveyor belt traveling helically about the drum from an entrance end of the drum to an exit end of the drum; wherein the drive element includes a vertically oriented rib extending radially from a surface of the drum; wherein the rib includes at least one drive face; wherein the conveyor belt includes at least one belt surface configured to engage the at least one drive face; wherein a height of the rib above of the surface of the cage bar cap varies along the length of the rib; wherein the rib further includes a chamfered surface configured to engage with a lateral portion of the conveyor belt, the chamfered surface angled facing partially in a rearward direction opposite the direction of belt travel about the drum; and wherein the chamfered surface extends vertically along the length of the rib and is thereby configured to facilitate engagement of the conveyor belt with the rib from a lateral direction.
8. The spiral conveyor system of claim 7, wherein the lateral portion of the conveyor belt includes a feature of an end link that is configured to engage the chamfered surface; and wherein the chamfered surface is configured to drive the end link in the rearward direction when engaged with the chamfered surface.
9. The spiral conveyor system of claim 8, wherein the feature of the end link that is configured to engage the chamfered surface is an engagement tab that extends laterally from the conveyor belt.
10. The spiral conveyor system of claim 8, wherein the conveyor belt is a buttonhead belt; wherein the drive surface of the rib and the chamfered surface of the rib converged to form a narrowed edge that is configured to fit between buttonheads of the buttonhead belt and the feature of the end link that is configured to engage the chamfered surface.
11. The spiral conveyor system of claim 10, wherein the narrowed edge includes a radially facing surface that extends between the drive surface of the rib and the chamfered surface of the rib.
12. The spiral conveyor system of claim 7, wherein at least a portion of the rib has a tapered height.
13. The spiral conveyor system of claim 7, wherein the rib is disposed flush with a trailing edge of the drive element.
14. A method of manufacturing a drive element for a spiral conveyor system, comprising: forming an elongate structure having a first surface configured to face radially inward on a spiral of the spiral conveyor system and a second surface opposite the first surface and configured to face radially outward on the spiral, the elongate structure further including a rib extending radially outward from the second surface of the elongate structure; and removing material from a portion of the elongate structure to thereby form a chamfered rib having a chamfered surface on the rib; wherein the chamfered surface extends vertically along the length of the rib and is thereby configured to facilitate engagement of a conveyor belt with the rib from a lateral direction.
15. The method of claim 14, wherein the chamfered rib is disposed flush with a trailing edge of the drive element.
16. The method of claim 15, wherein the chamfered surface is oriented toward the trailing edge of the drive element.
17. The method of claim 14, further including forming the chamfered rib to have a variable height with respect to a surface of the drive element.
18. The method of claim 17, wherein at least a portion of the chamfered rib has a tapered height.
19. The method of claim 14, wherein the chamfered rib has a maximum height at one end of the

rib.

20. The method of claim 14, wherein the height of the chamfered rib is configured to define a radius of a drum of a spiral conveyor system.
